

Category	Sub-Type	Algorithm	Purpose	Output Type	Common Use Case Example
Supervised ML	Regression	LinearRegression	Predict continuous values	Numeric	Predict house price, crime count
		DecisionTreeRegressor			Forecast number of sales
		RandomForestRegressor			Pollution levels, temperature
	Classification	LogisticRegression	Predict categories (labels)	Categorical (label)	Email spam, disease prediction
		DecisionTreeClassifier			Crime type classification
		RandomForestClassifier			Loan approval, image labels
		KNeighborsClassifier			Recommender systems
		SVM (Support Vector Machine)			Face detection, bioinformatics
		NaiveBayes			Text classification, sentiment analysis
		GradientBoostingClassifier			Fraud detection
Unsupervised ML	Clustering	KMeans	Group similar items	Group ID	Customer segmentation, crime hotspots
		DBSCAN			Anomaly detection
		AgglomerativeClustering			Hierarchical grouping
	Dimensionality Reduction	PCA (Principal Component Analysis)	Reduce number of features	Numeric	Visualization, speedup ML pipeline
		t-SNE, UMAP			High-dimensional data visualization
Reinforcement Learning	—	Q-Learning, DQN, SARSA	Learn through rewards	Policy / Action	Game AI, robotics, real-time decision-making

 Quick Rule of Thumb

Goal	Use This Type	Try This Algorithm
Predict a number	Regression	LinearRegression, RandomForestRegressor
Predict a label/class	Classification	LogisticRegression, SVM, DecisionTreeClassifier
Find hidden patterns/groups	Clustering	KMeans, DBSCAN
Reduce feature space	Dimensionality Reduction	PCA, t-SNE
Learn from environment feedback	Reinforcement Learning	Q-Learning, Deep Q Network (DQN)

 Supervised vs Unsupervised vs Reinforcement

Type	Input Provided	Learns From
Supervised Learning	Data + Labels	Labeled outcomes ($X \rightarrow y$)
Unsupervised Learning	Data only	Patterns or clusters in data
Reinforcement Learning	No fixed dataset	Rewards/penalties from environment